

The Classical Construction of Probability: A Categorical Reading

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Abstract

This note develops the classical construction of probability theory in categorical language, as preparation for identifying precisely where the construction assumes a measure and what structure that assumption provides. The goal is route-finding for the Observable Dynamics Program, which inverts this construction. We work through the category **Meas** of measurable spaces, the Giry monad $\mathcal{P}$ on **Meas**, its Kleisli category $\mathbf{Kl}(\mathcal{P})$ of Markov kernels, and the Kolmogorov extension theorem as a limit statement. The assumed measure appears as a choice of monad algebra — a free parameter of the classical construction that the program seeks to derive.

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1 The category of measurable spaces

Definition 1 (The category **Meas**). *The category **Meas** has:*

- Objects: measurable spaces $(X, \mathcal{X})$ — sets X equipped with a σ -algebra $\mathcal{X} \subseteq 2^X$.
- Morphisms: measurable functions $f : (X, \mathcal{X}) \rightarrow (Y, \mathcal{Y})$ — functions $f : X \rightarrow Y$ such that $f^{-1}(B) \in \mathcal{X}$ for every $B \in \mathcal{Y}$.
- Composition: ordinary function composition.
- Identity: the identity function id_X .

Remark 1 (Products and limits in **Meas**). **Meas** has all small limits. The product $(X \times Y, \mathcal{X} \otimes \mathcal{Y})$ carries the product σ -algebra generated by rectangles $A \times B$ with $A \in \mathcal{X}$, $B \in \mathcal{Y}$. Countably infinite products $\prod_n (X_n, \mathcal{X}_n)$ carry the product σ -algebra generated by cylinder sets — preimages of measurable sets under finite coordinate projections. This is the measurable structure on the path space $X^{\mathbb{Z}}$ used in the Observable Dynamics Program.

Remark 2 (What **Meas** does not capture). *The morphisms of **Meas** are deterministic: a measurable function sends each point to exactly one point. Stochastic processes, Markov chains, and conditional distributions require a richer notion of morphism — one that sends points to distributions rather than points. This is provided by the Kleisli category of the Giry monad, defined in Section 3.*

2 The Giry monad

Definition 2 (The Giry functor). *The Giry functor $\mathcal{P} : \mathbf{Meas} \rightarrow \mathbf{Meas}$ is defined as follows.*

- On objects: $\mathcal{P}(X, \mathcal{X})$ is the set of all probability measures on $(X, \mathcal{X})$, equipped with the evaluation σ -algebra generated by the maps

$$e_A : \mathcal{P}(X) \rightarrow [0, 1], \quad \mu \mapsto \mu(A),$$

for each $A \in \mathcal{X}$. We write $\mathcal{P}X$ for $\mathcal{P}(X, \mathcal{X})$ when the σ -algebra is clear.

- On morphisms: given a measurable function $f : X \rightarrow Y$, the map $\mathcal{P}f : \mathcal{P}X \rightarrow \mathcal{P}Y$ sends each measure μ to its pushforward $f_\# \mu$:

$$(\mathcal{P}f)(\mu)(B) = (f_\# \mu)(B) = \mu(f^{-1}(B)), \quad B \in \mathcal{Y}.$$

Proposition 1 (Functoriality of $\mathcal{P}$). *$\mathcal{P}$ is a functor: $\mathcal{P}(\text{id}_X) = \text{id}_{\mathcal{P}X}$ and $\mathcal{P}(g \circ f) = \mathcal{P}g \circ \mathcal{P}f$.*

Proof. Both follow immediately from the definition of pushforward. $\square$

Definition 3 (The Giry monad). *The Giry monad is the triple $(\mathcal{P}, \eta, \mu)$ where $\mathcal{P}$ is the Giry functor and:*

- Unit $\eta : \text{id}_{\mathbf{Meas}} \Rightarrow \mathcal{P}$: the natural transformation whose component at X is

$$\eta_X : X \rightarrow \mathcal{P}X, \quad x \mapsto \delta_x,$$

sending each point to the Dirac measure concentrated at that point.

- Multiplication $\mu : \mathcal{P}\mathcal{P} \Rightarrow \mathcal{P}$: the natural transformation whose component at X is

$$\mu_X : \mathcal{P}(\mathcal{P}X) \rightarrow \mathcal{P}X, \quad \mu_X(\Lambda)(A) = \int_{\mathcal{P}X} \nu(A) \Lambda(d\nu),$$

integrating a measure on measures down to a single measure.

These satisfy the monad axioms:

$$\mu_X \circ \eta_{\mathcal{P}X} = \text{id}_{\mathcal{P}X}, \quad \mu_X \circ \mathcal{P}\eta_X = \text{id}_{\mathcal{P}X}, \quad \mu_X \circ \mu_{\mathcal{P}X} = \mu_X \circ \mathcal{P}\mu_X.$$

Remark 3 (What the multiplication means). $\mathcal{P}(\mathcal{P}X)$ is the space of probability measures on probability measures — a distribution over possible probability distributions. This arises naturally in Bayesian inference: a prior over models is a measure on $\mathcal{P}X$. The multiplication μ_X collapses this to a single mixture measure by averaging. Classically, a prior is a free choice of element of $\mathcal{P}(\mathcal{P}X)$; the multiplication then yields the marginal distribution on X . The Observable Dynamics Program asks whether this choice can be forced rather than freely made.

Remark 4 (Monad algebras). A monad algebra for $(\mathcal{P}, \eta, \mu)$ is a measurable space X together with a measurable map $\alpha : \mathcal{P}X \rightarrow X$ satisfying:

$$\alpha \circ \eta_X = \text{id}_X, \quad \alpha \circ \mu_X = \alpha \circ \mathcal{P}\alpha.$$

Intuitively, α is an integration map — it evaluates a distribution on X to a point in X . Monad algebras for the Giry monad correspond to convex spaces: spaces closed under convex combinations. The classical probability space structure is a special case where α is evaluation of a fixed measure P .

3 The Kleisli category of Markov kernels

Definition 4 (The Kleisli category $\text{Kl}(\mathcal{P})$). *The Kleisli category $\text{Kl}(\mathcal{P})$ of the Giry monad has:*

- Objects: measurable spaces $(X, \mathcal{X})$ — the same objects as **Meas**.
- Morphisms: a morphism $K : X \rightarrow Y$ in $\text{Kl}(\mathcal{P})$ is a Markov kernel from X to Y — a measurable function $K : X \rightarrow \mathcal{P}Y$, i.e. a map that sends each point $x \in X$ to a probability measure $K(x, \cdot)$ on Y .
- Composition: given $K : X \rightarrow Y$ and $L : Y \rightarrow Z$, their composite $L \circ K : X \rightarrow Z$ is the kernel

$$(L \circ K)(x, C) = \int_Y L(y, C) K(x, dy), \quad C \in \mathcal{Z}.$$

This is the Chapman–Kolmogorov composition.

- Identity: $\text{id}_X = \eta_X : X \rightarrow \mathcal{P}X$, $x \mapsto \delta_x$.

Remark 5 (Deterministic maps embed into $\text{Kl}(\mathcal{P})$). *Every measurable function $f : X \rightarrow Y$ in **Meas** gives a Markov kernel $\hat{f} : X \rightarrow Y$ in $\text{Kl}(\mathcal{P})$ via $\hat{f}(x, \cdot) = \delta_{f(x)}$. This defines a functor $\text{Meas} \rightarrow \text{Kl}(\mathcal{P})$ that is the identity on objects and sends deterministic maps to their Dirac-kernel counterparts. In this sense $\text{Kl}(\mathcal{P})$ strictly extends **Meas** by allowing stochastic morphisms.*

Remark 6 (Stochastic processes in $\text{Kl}(\mathcal{P})$). *A (discrete-time) Markov chain on X is a morphism $T : X \rightarrow X$ in $\text{Kl}(\mathcal{P})$ — a transition kernel. A stochastic process is a diagram in $\text{Kl}(\mathcal{P})$. The Chapman–Kolmogorov identity (that n -step transition kernels compose correctly) is exactly the associativity of composition in $\text{Kl}(\mathcal{P})$. This makes $\text{Kl}(\mathcal{P})$ the natural ambient category for the dynamics studied in the Observable Dynamics Program.*

Example 1 (Refinement maps as Kleisli morphisms). *In the Observable Dynamics Program, a refinement map $\pi : O_j \rightarrow O_i$ between query outcome spaces is a deterministic measurable function. It embeds into $\text{Kl}(\mathcal{P})$ as $\hat{\pi}(o, \cdot) = \delta_{\pi(o)}$. The compatible marginals condition $(\pi_{ij})_{\#} \nu_j = \nu_i$ says exactly that $\hat{\pi}_{ij}$ intertwines the marginals ν_j and ν_i — a commutativity condition in $\text{Kl}(\mathcal{P})$.*

4 Probability spaces as monad algebras

Definition 5 (Probability space as algebra). *A probability space $(X, \mathcal{X}, P)$ can be viewed as the measurable space $(X, \mathcal{X})$ together with the choice of a point $P \in \mathcal{P}X$ — equivalently, a measurable map*

$$P : 1 \rightarrow \mathcal{P}X$$

from the terminal object $1 = (\{\star\}, \{\emptyset, \{\star\}\})$ to $\mathcal{P}X$. This is the free parameter of the classical construction: nothing in the structure of $(X, \mathcal{X})$ forces or determines P .

Remark 7 (The free choice). *This is the precise categorical location of the classical assumption. The measurable space X carries no preferred measure. Kolmogorov’s axioms do not say where the measure comes from — they only say what properties it must have once chosen. The choice of P is epistemologically a prior or model assumption; it is not derived from the structure of observations.*

The Observable Dynamics Program asks: can P be replaced by a canonical construction from observable data, so that the map $1 \rightarrow \mathcal{P}X$ is not freely chosen but forced by coherence constraints on a query system?

Remark 8 (Conditional distributions and disintegration). *Given a probability space $(X \times Y, P)$ and a measurable function $f : X \times Y \rightarrow X$, a regular conditional distribution is a Markov kernel $K : X \rightarrow Y$ in $\text{Kl}(\mathcal{P})$ such that*

$$P(A \times B) = \int_A K(x, B) P_X(dx)$$

where $P_X = (\mathcal{P}f)(P)$ is the marginal on X . This is the disintegration of P over f . Its existence (under standard Borel hypotheses) is the Rokhlin disintegration theorem used in Paper 2 to construct the predictive kernel.

Disintegration shows that the Kleisli morphism K is derived from the algebra structure P — not freely chosen. In the program, the minimal predictive query reverses this: K is derived from the query structure, and P is the algebra structure it forces.

5 Projective systems and the Kolmogorov extension theorem

Definition 6 (Projective system in **Meas**). *A projective system in **Meas** is a functor $F : I^{\text{op}} \rightarrow \mathbf{Meas}$ from a small directed category I^{op} to **Meas**. Concretely: a collection of measurable spaces $\{(X_i, \mathcal{X}_i)\}_{i \in I}$ with measurable maps $\pi_{ij} : X_j \rightarrow X_i$ for each $i \leq j$, satisfying:*

$$\pi_{ii} = \text{id}_{X_i}, \quad \pi_{ij} \circ \pi_{jk} = \pi_{ik} \quad (i \leq j \leq k).$$

The projective limit (inverse limit) is the measurable space

$$\varprojlim X_i = \left\{ (x_i)_{i \in I} \in \prod_i X_i \mid \pi_{ij}(x_j) = x_i \text{ for all } i \leq j \right\}$$

with the subspace σ -algebra from the product.

Remark 9 (Query systems as projective systems). *The query system $(\mathcal{Q}, \preceq)$ of the Observable Dynamics Program is exactly a projective system in **Meas**: the outcome spaces $\{O_i\}$ are the objects and the refinement maps $\{\pi_{ij}\}$ are the morphisms. The realization space $\Omega = \varprojlim O_i$ is the projective limit. The Observable Dynamics Program therefore begins with the projective limit already in place — it is asking for the probability measure on this limit.*

Definition 7 (Compatible family of measures). *Given a projective system $\{X_i, \pi_{ij}\}$, a compatible family of probability measures is a collection $\{\nu_i \in \mathcal{P}X_i\}_{i \in I}$ satisfying*

$$(\pi_{ij})_{\#} \nu_j = \nu_i \quad \text{for all } i \leq j.$$

This is a cocone over the projective system in $\text{Kl}(\mathcal{P})$: each ν_i is consistent with the morphisms of the diagram.

Theorem 1 (Kolmogorov extension theorem — classical form). *Let $\{(X_i, \mathcal{X}_i), \pi_{ij}\}_{i \in I}$ be a projective system in **Meas** where each X_i is a Polish space with its Borel σ -algebra, and let $\{\nu_i\}$ be a compatible family of probability measures. Then there exists a unique probability measure P on the projective limit $\Omega = \varprojlim X_i$ such that $(\text{pr}_i)_{\#} P = \nu_i$ for all $i \in I$, where $\text{pr}_i : \Omega \rightarrow X_i$ is the projection.*

Remark 10 (Categorical reading). *The Kolmogorov extension theorem says that the functor*

$$\Phi : \mathcal{P}(\Omega) \rightarrow \{\text{compatible families } \{\nu_i\}\}$$

sending P to its marginals $\{(\text{pr}_i)_{\#} P\}$ is an isomorphism (bijection) under Polish hypotheses. The inverse of Φ is the extension construction. Categorically, Φ is the counit of an adjunction between the category of probability spaces over Ω and the category of compatible marginal systems.

The Polish hypothesis is what forces the adjunction to exist. Without it, Φ is injective (uniqueness: the Observational Determination Theorem, Paper 1) but not surjective (existence may fail).

The Observable Dynamics Program replaces the Polish hypothesis with sequential upper-directedness and realizability, obtaining existence by a different route — a Carathéodory extension argument rather than a topological compactness argument.

6 The assumed measure: a summary

We can now state precisely where the classical construction assumes a measure and what categorical structure that assumption provides.

Step	Construction	Categorical object
1	Sample space $(X, \mathcal{X})$	Object of Meas
2	Giry monad $\mathcal{P}$	Monad on Meas
3	Choose measure $P \in \mathcal{P}X$	Free point $1 \rightarrow \mathcal{P}X$
4	Pushforward along f	Functorial action $\mathcal{P}f$
5	Markov kernels	Morphisms in $\mathbf{Kl}(\mathcal{P})$
6	Compatible marginals	Cocone in $\mathbf{Kl}(\mathcal{P})$
7	Kolmogorov extension	Limit/colimit in Meas (Polish case)

Step 3 is the free parameter. Steps 1–2 and 4–5 are structural (forced by the category). Steps 6–7 are consequences of Step 3 plus the structural machinery.

The Observable Dynamics Program begins at Step 6 and asks whether Step 3 can be derived as a consequence — whether the compatible cocone in $\mathbf{Kl}(\mathcal{P})$ forces a canonical point $1 \rightarrow \mathcal{P}\Omega$ without a free choice. The extension theorems of Paper 1 give conditions under which it can.

Remark 11 (The adjunction statement of Paper 1). *The main theorem of Paper 1 can be stated as: under sequential upper-directedness and realizability, the functor Φ above is an isomorphism. Making this into a precise adjunction statement — identifying the two categories and the unit/counit — is the next step of the categorical development. See the companion note on the program’s picture.*

7 Mathlib availability

See `mathlib_gaps.md` (this directory) for the consolidated audit of what Mathlib 4 has and lacks for this picture — Giry monad, Kleisli, Markov kernels present; projective limits in **Meas** and the Kolmogorov extension theorem absent (formalization gaps, not mathematical ones). The classical construction here builds on `GiryMonad` and `Kleisli` and connects to the in-house `QuerySystem.lean` via the embedding $\mathbf{Meas} \hookrightarrow \mathbf{Kl}(\mathcal{P})$.